WORKSHEET

Chapter 1: Vectors in polar form

1 Convert the following vectors to polar form.

a $(1, 3, 2)$

b $(2, 1, 4)$

c $\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$

d $\begin{pmatrix} 5 \\ -2 \\ 3 \end{pmatrix}$

e $(10, 4, -6)$

f $(3, 1, -5)$

© Cengage Learning Australia Pty Ltd 2014

g $\begin{pmatrix} 4 \\ -3 \\ -1 \end{pmatrix}$

h $(-3, 5, 8)$

i $(5, -12, 7)$

2 Convert the following vectors to component form.

a $(4, 30°, 60°)$

b $\left(6, \frac{\pi}{4}, \frac{\pi}{4}\right)$

c 8 at azimuth $\frac{\pi}{3}$ and altitude $\frac{\pi}{6}$

d 3 at azimuth $\frac{5\pi}{6}$ and altitude $\frac{\pi}{3}$

e $(7, 317°, 32°)$

© Cengage Learning Australia Pty Ltd 2014

f $(10, 15°, -30°)$

g $(8, 320°, -75°)$

h $(5, 32°, 63°)$

i $(16, 26°, 87°)$

3 If $\mathbf{a} = (3.742, 63.4°, -53.3°)$ and $\mathbf{b} = (5.477, 291.8°, 10.5°)$, find the resultant of $\mathbf{a}$ and $\mathbf{b}$. Express the answer in polar form.

4 Given that $\mathbf{p} = (7.55, 315°, 68°)$ and $\mathbf{q} = (11.18, 309.8°, 45.7°)$, find $\mathbf{p} + \mathbf{q}$. Express the answer in polar form.

5 If $\mathbf{a} = (4.583, 26.6°, 60.8°)$ and $\mathbf{b} = (3.742, 341.6°, 32.3°)$, find each of the following. Express the answer in polar form.

a $\mathbf{a} + \mathbf{b}$

© Cengage Learning Australia Pty Ltd 2014

b $\mathbf{a} - \mathbf{b}$

c $\mathbf{a} + 2\mathbf{b}$

d $3\mathbf{a} - 4\mathbf{b}$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $(3.742, 71.6°, 32.3°)$

b $(4.583, 26.6°, 60.8°)$

c $(3.741, 26.6°, -53.3°)$

d $(6.164, 338.2.8°, 29.1°)$

e $(12.329, 21.8°, -29.1°)$

f $(5.916, 18.4°, -57.7°)$

g $(5.099, 323.1°, -11.3°)$

h $(9.899, 301°, 53.9°)$

i $(14.765, 292.6°, 28.3°)$

2 **a** $(1.732, 1, 3.464)$

b $(3, 3, 4.243)$

c $(-3.464, -6, 4)$

d $(1.299, -0.75, 2.598)$

e $(-4.342, 4.049, 3.709)$

f $(-8.365, -2.241, -5)$

g $(-1.586, 1.331, -7.727)$

h $(-1.925, -1.203, 4.455)$

i $(-0.753, -0.367, 15.978)$

3 $(4.69, 315°, -25.2°)$

4 $(15.811, 306.9°, 71.6°)$

5 **a** $(7.81, 0°, 50.2°)$

b $(3, 296.6°, 41.8°)$

c $(11.358, 352.9°, 44.8°)$

d $(10.05, 310.6°, 23.4°)$

© Cengage Learning Australia Pty Ltd 2014